ECG Dataset Compatibility Study — Metadata Only

Purpose: compare the structure and acquisition attributes of eleven ECG datasets before designing an acute STEMI / NSTEMI / non-MI study. All code, output tables and figure captions are English.

Start here: stop any old full-download or waveform-analysis cell. Open this notebook in Google Colab and run Sections **1–6 in order**. The default study needs **no ECG files, AWS credentials, Google Drive mount, or complete MIMIC download**. A CPU runtime is sufficient. Your existing datasets are not modified.

This notebook replaces the earlier inventory / waveform-feature workflow for this question. It does not call `subset.inventory`, `run_full_downloads`, or the old audit engine. It does not read `.dat` / `.mat` sample arrays, calculate ECG morphology, train a model, or automatically create clinical labels.

Sections **7–9 are optional** local metadata/header/schema inspections. They require explicit paths but no full folder scan. Sections **10–11** save results and describe reconnection. Documentary conclusions and locally observed metadata remain separate.

The proposed default model input is standard twelve leads, 10 seconds, 500 Hz. Edit the target before running if the AI team uses a different specification. A common sampling rate alone is insufficient to approve pooling.

1. Check dependencies

Run once per new runtime. Only missing packages are installed. Most standard Colab runtimes already contain the four packages needed for the documentary analysis. Optional header/HDF5 dependencies are installed only when those sections are enabled. Exact loaded versions are recorded with the results.

```
import importlib.util, subprocess, sys, time
started = time.perf_counter()
requirements = {'numpy': 'numpy>=1.26,<3', 'pandas': 'pandas>=2.2,<
                'scipy': 'scipy>=1.12,<2', 'matplotlib': 'matplotli
missing = [package for module, package in requirements.items()
           if importlib.util.find_spec(module) is None]
if missing:
    subprocess.check_call([sys.executable, '-m', 'pip', 'install',
print(f'Dependencies ready in {time.perf_counter() - started:.1f} s
```

```
Dependencies ready in 0.0 seconds. CPU is sufficient.
```

2. Load the small standalone engine and evidence registry

The notebook is self-contained. Compressed payloads use short lines to avoid the earlier giant-source-cell rendering problem. Loading only reconstructs two small files on Colab's local disk and imports the module; it does not inspect your data or access the network. The readable `.py` and `.json` are also supplied in the package.

```
from pathlib import Path
import base64, hashlib, importlib.util, json, zlib
from IPython.display import display, Markdown, Image
import pandas as pd
WORK = Path('/content/ecg_metadata_compatibility')
ENGINE_DIR = WORK / 'engine'
ENGINE_DIR.mkdir(parents=True, exist_ok=True)
```

```
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```

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```

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'wa8dVQk1W5QsJ6ZMBFo8DfM1JG3iUZGbg7fa9RVJvDg+m/6hiAf0FWrYCuMK5B
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'0jxjWhSJTd29eSgZCS7Fa0OE+8hVrmucn2NBDEW8RmY7X8W+zdwWqaN9GMwXMW
'8DR6A/iygKAwaJBHj0+DJSB/ts42Wa00IVVKqOD89Pzbs1MCFFaBcQABQ6Hlcn
'ZbAQhtfRynu0/HLJDgjcMXjEaifG4IA56C1U2uqjt2OuOefUDEZDswS2Ns4Mii
'pUwvweqTM748tVlQeZH/cH46mED02aqA6egsDKmAvkhd/CUaBQU353Bz+voTFu
```

```
    'cOJeI1ssaEHRQ+KBiALWKvhhUhZb4qrFbNEeunycGWV95Vlf7bHwv6EsvYeUuN
    'daAPiBPP4tUWMPTlWBQYBuPam3BrQdOaJ6hfwE9uYJm229FMNLckFe3PilKwas
    'tcrTHrqQUCHZJd4wMBL802+hBOs50OcrEk01teILssA/BxIAqUWN2ourRXiTa1
    'luiQEaVRVoiw8sqiqANt/RS8yNyJvnES1geeGquKZeqPTwODzTp4j6X5v//0/E
    'ihmCr6/NwJw9iQR5RAfQhrLqcSGj0BQkYi55KBQPhSdcZMseV94fUuGO7Ssj2k
    'NvY6KbucQGLuVLqwM1VF7bzvauNMDdmIiH4HWgwB5WxujdIO1C04ZU0mr9IJ1b
    'HNq8SwVohOJRhqsZA32iE90YSobm3QcHY7TpxS2CBiS1e6NIH/WS0/bO6SoCCI
    'NVVKBECOysDcOc70mMqwo4Cyt1vpJ41NXWQpLFISb2tzhV5HH2jULGq7EYpix2
    'KfobfaGy+YGwFyuuSzevKpxIR819X/YFplIqZ+zK9z60P/KrAz3GyWJ2TcDRrq
    '3hkymCuY72vWctiK8/5GWW5k9s0xvaunD6c9am9T9yIRA9uRxuFyOzCTT/1K6C
    'ByGUBEkTbWTFN8LFgpiOmg+qubvBarUXtnHJf9h1AR4ci4sh+E9xytyA/7AGiw
    'GX2/P1GReS0UfW8oIDvOdLSOg1K2WHIs0yynEILuUlhQHUzFVyu1sDsDg2A43X
    'oDHNDaxSYEmMGL/R9yxDeXFQRRwcCBUxDTJRVdlJFWdgT1pfVGO8pBt/GeXNEr
    'S5vV+gJN0HVRJEoHomCckfx03E/DDUJEpcN0s633GoRxlwCWi60mhqQqeIV7SC
    'c4xzwo30DANsm5QcR5+Zu+L74C9+T4mvvLV/6N7oHt+FKzBs1giCOXHKSAYPfZ
    '/ljiDB5xLzdHE2KvtE0mENRsvdrxw5lzLUzX8QZsHSUvjxgFYv38e/9BnZQZhU
    'H3nl7nuyUbJ83PjKsEncc0lMW8A/hBYX4bFwUfDZJsffz08bGeildpLXKh52ly
    'LmLw2BUoyz9UXeEIEZB2WhiMwv9Bj2fwRryigFfd3jIJoOjA9XfsfYYOwp/AQ/
    'VUSFlW3RPUHgA5zS+BE3PbyPloOs0JZDVnJpYpjRwTpF1AKzOSgX0AO/bKdeH5
    'ivx4BXd/VJeMH7KTrXv37P9hqvTfBAOdDskP1pionAzpeYtUDLkCtIxXuPaW/g
    'cnebkrfAZBsQSI5p5ffN78NNNqyg5KS12maJukmRlYn2HS1zls2QZWw/oIPoDw
    '6MOdDfOlh3nDjrWlxbGNO0yH+L6pK6tnNZH/tegRrgPBbDdpyQxey6+PwK5xa+
    'ursG8dk7nH9MojKgUi536lLkL7uFU27NgDmPrUYTlz1qm6D6i65+AO9xLzudY9
    'vpFLZ45D2Wa7co1fg9CviqgoSxRTsr71LLHZaE+UhJpzaxnbcN8Cl/pkanCGgx
    'orwyeVPGdHTSbZttMOF+HdqGhEb8W+Q4WIUJ9yVUn1MdcX4RZ74nEhx9VDtykT
    'qdyM1CQtUel0RVn9YO5PITKQTNn42gAHMROurx1tBi6JUhfF27WXdhVbS9ASDh
    'sn1KBvR3HBKoBCRmMnWgRqPJUgPTwcVWULHuPpBIazbJcMfWTRQRb/lE1Adqsr
    'V4fO1acWQLv6Q+8sYYBthaViXzkEqX9IurPFtT0sTymPjVb+IYzZ07iIX/LdDI
    'Qe20e23qLm1uOuv3/yolftyOb2adxnwds4+dRNkVSzUiMFU58ftDa9Fck8bL6F
    'A3EpX903u0ufRKefhHodh2nHjFk5KHW4i7oB0dgu//3qlzeIJ523pS97opPWvd
    'jNl/h7NjFmuOru+ma+2PqJU8Hfd8a5WuGpA3OCK5h06ttwCL+oFbmOzisKmri5
    'hVQnNmKE4yMEDDYpeu6qrrfVxckJMs7QI578wbV9qqPVf3v7ysnMvtu8Pw5nGC
    '0F/77LxBhYYymvRDE+pcXePEMn+k0FjaZHe4cNzhLgPwM73hWQjb8GNKpyL9CH
    'pTmn4aEKx2TjeWGTlnq+pfiNTo40JTqa3o+ZGLU/MFnkbx81RT5CS0fBUXNjUD
)
payload = zlib.decompress(base64.b64decode(PACKED))
assert hashlib.sha256(payload).hexdigest() == '4054956e2849d782c6fd
(ENGINE_DIR / 'ecg_metadata_compatibility.py').write_bytes(payload)
print('ecg_metadata_compatibility.py', 'loaded:', len(payload), 'by
```

```
ecg_metadata_compatibility.py loaded: 40689 bytes
```

```
PACKED = (
    'eNrtnelz3DaywL/nr0DNq61K6mlOS5YVfZJHiuXdleKy7Gyy+1JTGBIzREwSsw
    'jeOH5n++YayjvUBEfLIUiZYq7nzPOsPeoHOAt7ws4SlcnPg8FXhnNBg97w4H8N
    'iZxmqfBZcYvNEvHvTMTein17+fG7UxarlPEsVRFk5fEwXDGVyLmMeci49+9Mac
```

```
'ETNxBy97MoXfhU75NJQ6EH6RSTwJBTeV26iHF/A4FiHzVBanxbPwduzzxJ8MR/
'wfiSyxDyLiuHtyYq8UViqwQ/5C+Yd/VGDapis5nKMxSh8LCkwjcCKNI1lxPli8
'DaEyvV69XBSa3UIRWSI8LDW09Nrj4apIxC90KZLxxEpVeRlmB6/ANXgpqtIonj
'yc3knfAP2JInEpsHtNK8UimIThW0jNVBYUrwA77C7HWr8MLqiW0tvyxgkYQtHy
'QNNoL5ELyOLA9BzpQ4LYb5gHHcH8cKuSD7NQ3RaZRDzOZtxLs8Sq1N9idRuz9a
'CSOqEJM8W+tv9h4z9/K0YtB0L5Cg/XqlUxFhvTyeFKUHK5IUKc9kEt1Cv7DitD
'V4LHaHWgE7JQRjI1jawxhbygVRlKhV8EK42inGSxTE0yb/IrzBicFBSI4T0mZz
'VLEiZ2Zzzl+fc+gz3PcmC5FMoqWR+PnYPsPgGot+shrGQU1itDhQMr1u33ZVD5
'WGg+By1MV4vSYAVyHiy41rkRXtcFvNXFe8wDCz+blTZO3d7zBtypewEK4gU1j5
'DGQDIsFqLYSHGmAbEISYKK27WmWJhx04WkDtddWai0RAyigV7BKblgdkrhLoQy
'WOG40nCeJHw1CflKZeaNG6tB5jLjd+gZip7LoTG7ea08+B16OYjWvlmoJNolKA
'0CrV4sEJqGGS2vZKM7RPVRrmFlsJntS8J2K//i24sfEODkrqO7sxD+AxNVp+eE
'Ne8P7lvAWMSYtvVUweLgHbaE96ZZyCu52jcSMQPHGXuisAmLzRdzxwq9Qc7jqu
'SKCbljoqB09gREO+mECNROUOQF680l9tvB08AKKFKuevoIFKwGiK0rWAeRGxFm
'Q0K/m7Gk7W5rg8lFOr0Ly1/hAuid7cxv3r3s/vx3BuPU3rO8KHA7S4wvCtJ0ob
'vPp74SfBPXi1+V29vno97r7+qXsxfsWK0XHzPE3CXbnsCm+OOW/lC+0DOh6blq
'lL/qlhn1NsuJyd1XpknekWrLtlUiI1TmF4as8n9NBVXm2jVFqmskGEbEvLYw16
'pi1uWkhoxC7Y3GCVBpHkppDbLaoXQW0R/y4zBg3Jwx67eXOJpomvDejYt1Dr0X
'x114tzsYHg6edY82SwMqf18vYufVWNNU5OG9yZ/WikDGnjX/u/m+vh6fvX33qF
'NHh+b+YfRax8ZXK2HkP3i1d2cpxsDAV2Mz/L0gCN6fqYoRyO8b2dZC7TIJvaxc
'0Yv8vJylcUVVA1eN1vVf5lZZUGm8LTTkBIxtWRKcEFSWtrpeWYOO7e39fCCujw
'PXx0jxMz2sRSFu56Ui9Am5Hp6zARqCpb3rMuZe3arxuvooOHjmiGY+t9E41FHz
'/75FjVpJxVkY7hdVFn9Av75PNPfW/fNreA2Ofn1eJGM7LTILSjjKyIuLw8devT
'LlS/4ueqNLZWMV0Vw846bL15ddkT7ywtk4juF9HmoHtDQnYJveO0o97aIrhv3C
'EBf4X91fKccs2OYWlqvDgbX9NmdHRLU7h/UW6I0Wma/iVSQ/4t6CebOPuySorh
'2f12tljr5R395LKQNreN6yV0aTb7zM4tbhObNe4ph6xlLFyW3e6e+T2jeKfHXm
'VRDbkE+9970pQBGYw7PLjzk+cco4SjGGvNDB4A2Ups5t+1RYyIRHgl2MX2mX3X
'VeWVAE2KV04vN1pcz10N2YEP68f8Jy9OXJ5VbrOW9wjp5LnLy4PbcOk960EWt+
'GVmO3yBu2nFeTuuUdL2okMqF0Tpyy6csyKDPMrgnfW7Pf5kzajvnrNxcdd1Bx+
'ynkgTieFLgQEilrCzvmksaDsNYPnzVkpIzanDcDm8YL6BhiP2ctf2GFvcIrHw+
'/38xY6bkeMaX48ZBz9NLvKq9BQ6jU8ON6eJqpd/9Af4x/w4nQdILBN//AGSw9s
'LXZnfS60U/xCNGW7dIojkxOutdC6OBJbnd9Bz1IlwvCYmYReLWNfLGDsjAdt8j
'rPRk5aHgXme3LJifPWeLUyKj0uXJNJtDkAjR9UI8JaezhUiWUm9g5KgLJoH3Wq
'fCb89e0JKB6apaoeuzSHJswZRVBgAf0zKFMYDQ9ePDtmvpzN8EzjLFFR2QcNLt
'0O0ZIwLQCUD/qlWlHQB6myVEADoB6ASgtxhAb7PxaAmA/nWI6E8H0NvtqFsCoL
'ZWMY5kCGcrFz+ZWIRYJzTzOS2CcRx8wf0etErz+YXsfgCfAkxoZCfe2Woa/87X
'GjLmDkiCOHLiyIkjJ46cOPJHcuQ2QBkG+uOhzoOUYQ6pYshAuzzoagki3mYRES
'CwQmi10J32z1jzG6beKy8JrC5aPB4AWJ6RNw+WgwPHF59e4z4PJTZjjCZCnYrQ
'WgD0jvPr/fsZ9ReDweDZkcsAQxNI/YpjPF6R09cM907ySPvxB1xULD5JkENoHe
'L+VrIgnHAr86IRY8sbzxWqBuNxdrG8LuuZ2ECp4ynWb+yiyRF0vhGj9f8QUOYr
'4nEvotBZp81DE9T9rWWN7XkXi9UWofbR6pZ9v5Bx19iFL0K/t062++n3dda4X1
'o1+erIpnb/kSgz9Fpm2NNSqTMVR7f/Ez/Okvbt6//OvF+F1f37x7f/5L3/z7UC
'PKslDJ9emofcahvF5v5vGc5D7Rec1tWfhzBPZb4CMx0XtPwq5+Rv3l1cve5f2/
'PK4CQeRj+HyAHgh4VM1FLGS66rFrtfE1JbSVMoYBBH72xcD1+YbuXqw+Nt9F2A
'/+rVpR2UfdulRKQ9kfZE2reYtG+7AWkJbf/1iOlPJ+7b77ibUPcOWKA2kvVtV5
'CUEnBJ0QdELQ/yAE3QWFaACSXz09SL77bfWDr2uG2oQ2N8vymt2xfBGEfQt3vi
'XVgjaUSUO7PivJ/tfnZ4MjgiT/RHBCm//vHq4pyN32FT5RX1WWA+Saddd1MNiW
'IJK8iBMuYy/MQJOZJfgMZDgO+AKMf/cm4OruC3xatPV97wE4dRWIfT03I9YC2C
```

```
        'FaIn+NXcYeztxfjHt+eT1+dFsPD1SzA1Z+x/8MsN37OtyeM+9tmdaN5jHNNomy
        'NpSxZrkFPWWvLtjV+5sLZIqRnF4kCiqN9QbN90Aw2oT9tuDxXr7Ys6ZnordNT0
        'MRFzTMwxMcfEHBNzTMwxMcfEHBNzTMwxMcfEHBNzTMwxMcfEHBNzTMwxMcfEHB
        'E3NMzDExx5/JHG9zp0Qfu0gfz4VK5pLvQsev7A02HHWxPYk5JuaYmGNijok5Ju
        'oo6JOibqmKhjoo6JOibqmKhjoo6JOibqmKhjoo6JOibqmKhjEu1jqOP3N2fEHB
        'c5yzsPmzjHuJ0rriX5F2rPBCCyLrU/ZBiEXRBxEcXIqYo9oRfUz08Zejj3PulK
        'gjTFmlojBNxj4zc3Y1bwHCzheAfMWKGkeZnMm8XpB6ul++MdL7Q3GQ2GL2oCHW
        'NqGNv9z6FOHGhBsTbky4MeHGhBsTbky4MeHGhBsTbky4MeHGhBsTbky4MeHGhB
        'Jt6YeGPijYk3Jt6YeOM/MNpxQZ0ScewmcSzu0oTXI8fdi817xBwTc0zMMTHHxB
        'xBwTc0zMMTHHxBwTc0zMMTHHxBwTc0zMMTHHxBwTc0zMMTHHxBwTc/zVMMfPDc
        'zkHHehHs0sY3AchDxXP2BmySjD0JynCp9EKmPKzlj3+Qcx3wBLMuq3h0fHzyYj
        'Y6KOH08dD931wc2A4+ckoHaQxm320cQYt5Yxbq/aNKGLtypzJfyLu/ur45Rxag
        'TAOcufMmFyOLfmIGwtF5GafQDNMVK8XjrhiJhyYemnho4qGJhyYemnjoz+KhL8
        'bKPcQY1qBjuPBsMTEtJeDHo0GDm8rdAIkO68sahiVwdyhl4UX3PYnjehqUdHx8
        'YSUjUPqBoHRhpSavz93VmiYotQuLOw/gm0uyssDE0QDBmEp6oUBPx7uZ/kJoc8
        'wAk1s5jzr/01vvkIWebhCGaIBbfQBGZe27Rdy7izMe6zBShq4g7+vA9DXqTTSc
        '8XDw5Mgx1Pi/ADl2WlCfgpGPnJVMO2Dk9sqHYGSCkR8BIzvdZ5phyk7x2vtwZK
        'YmfUBCq+hom2p6JIJGZjcJGoVBnZOCs2gowJMm4nZNzePtMEMr404XvY2fmYIU
        'IRKZSGQikYlEJhL50ZGZz58+MnOLXWsDDNmJeK/7+eJXAvQoXrmrJ3sJ4/exXI
        '/nd4ikTRlR+GFVN0ZYquvC6Go8MTd/fim9DA45yHYjqLIp7IpwfM2y6wT7K/lc
        'r1AfiBHnxKK78moUlfnawIBb0ZfNcAp1L0/ilPnKRGFdKBUyrjd4Sisclzt7E1
        '0fX/+2W6H1fe5F8NuZxbhMFgmAdf1vDjAOxuFXu5vOLb0MsFdsfSWxGC4v+vnS
        'w32UxAEDibBporjPrl6zvFYwZoG/2M14nJPDBZNs/MkTs8fIDy/AroLymdjr0M
        'zdgQH+QGEvPXteUU6lQ0FUtVCiKCeqLwktR8VwFah701QzCRsCkkcyv9NCiEgY
        '8ZuhHjjh2QGmX1+Pz96+e3JQenR0/MTdOK8gRWem6Mxft660g4lutYgIiyYsml
        '8hhCLXfYhFU/EKsOFEICTmtNE6raHcx1LyDtBDVDwDMBz20EnlttRJswz3ZRra
        'EThbiIQ5LlwKt9wUfqZwywQ5E+RMkDNBzgQ5/6mQc6vHE8Q5N+Kc32bQmbjbK2
        'GnIpzQlbBJfKBfxuKJYixI/Nm3HNTgGclChFMW6MHzsgiUZg8bikiXU2hbRYwD
        '56lrkZbxj79IqOP2iXE/PmzLazjh14NhAQjjjzC+KH7kaYKQ8HTLQ3wWF/w3IR
        'iNWDG2K3MvbVbe6NiqSZCOVcTmWIhx3RbV2a3eSuryJc2rehjf/EGMUYuXl4tI
        'ubjt4ZOTK1hJgm0Jtv2qNaUdqG2LBUSgLYG2BNo+HrQ9dtd0NGRsBySh/Xztgv
        'xN4Se0vs7Wexty7svn4ar3VACsTBEgdLHCxxsMTBEgf7WRxs562ws1gfD5wcDg
        'PC/jIkENV9VxFrPHkWY+ft8T2gIKkn4RYMMzC6JVnVv6IZH9XO7l+Q9HXfxurs
        '71M878uEf5Shywt4RPM2oXlHA5cXw5uBviAjh1d5iQF+AAP87PDo+PjE4a37Jh
        'sWYwwMs0S7kXrDwclfKO63IlFrh5KOLiOemfMhxGwg+GubvlS4Hjc/abkrHDGt
        'mbsMhOvDXHFp2VhxZ0KZutxnHwAU5yBmgRHnr7omvP0YsSVcDUZs5oU9Ty/LXy
        '47k21+zePUwlv9tMY2jSYL1ebx96vMZpFUUp6vFJKvlNVpg/GJjPoXvNpAh9XQ
        'PGcqDleMz1K4nAovsNwyssrcwwWTLwMpo1z7uMwEdt1SFKxgTjCucZgZxhoBE/
        'vwp0GVDfLE6EVuFS+EUoYvj3129+/+b/AY4fX54='
    )
    payload = zlib.decompress(base64.b64decode(PACKED))
    assert hashlib.sha256(payload).hexdigest() == '3beed05197f790c4b5a1
    (ENGINE_DIR / 'dataset_registry.json').write_bytes(payload)
    print('dataset_registry.json', 'loaded:', len(payload), 'bytes')
```

```
dataset_registry.json loaded: 130649 bytes
```

```
spec = importlib.util.spec_from_file_location('ecg_metadata_compati
compat = importlib.util.module_from_spec(spec)
sys.modules[spec.name] = compat
spec.loader.exec_module(compat)
REGISTRY = compat.load_registry(ENGINE_DIR / 'dataset_registry.jso
print('Engine ready:', compat.VERSION, '| Dataset profiles:', len(R
```

```
Engine ready: 1.0.0 | Dataset profiles: 11
```

3. Configure the study and review source evidence

The default is a documentary assessment of all eleven selected distributions. Missing information is retained as `Unknown`. The dataset names in the earlier screenshot mix release editions: native PTB-XL 1.0.3 has 21,799 ECGs and native INCART has 75 records; the challenge INCART subset has 74. Always report the exact release.

A sampling frequency is measured in Hz; do not count “Hz” and “sampling rate” as separate features. Original acquisition frequency, released frequency, and filter cutoff frequencies are different attributes.

Results are written to Colab's **local runtime disk first** to avoid many small Drive writes. Section 10 can save a single results ZIP to Drive. No source-data path is needed here.

```
RUN_NAME = 'metadata_compatibility_v1'
OUT = WORK / 'results' / RUN_NAME
TARGET = {'sampling_hz': 500, 'duration_s': 10,
          'require_standard_12': True, 'recording_context': 'diagnos
OUT.mkdir(parents=True, exist_ok=True)
compat.export_review_template(REGISTRY, OUT / 'evidence_review_templ
PROFILE = compat.profiles(REGISTRY)
display(PROFILE[['name', 'release', 'sampling_hz', 'n_leads', 'forma
print('Review template:', OUT / 'evidence_review_template.csv')
```

	name	release	sampling_hz	n_leads	format	manufacturer
0	PTB-XL	1.0.3 / records500	500	12	WFDB-DAT	Schiller AG
1	MIMIC-IV-ECG	1.0	500	12	WFDB-DAT	Burdick/Spacelabs; Philips; General Electric
2	Ningbo	Challenge 2021 / 1.0.3	500	12	WFDB-MAT	Unknown
3	Chapman-Shaoxing	Challenge 2021 / 1.0.3	500	12	WFDB-MAT	Unknown
4	Georgia 12-lead	Challenge 2021 / 1.0.3	500	12	WFDB-MAT	Unknown
5	CPSC 2018	Challenge 2021 /	500	12	WFDB-MAT	Unknown

Optional: add reviewed evidence before analysis

Leave disabled for the initial run. To resolve an unknown, fill the exported CSV with a typed JSON value (for example `500`, `true`, `"mV"`, or `["Manufacturer"]`), an exact source URL and locator, reviewer and review date. Keep release and scope unchanged. A single sampled header cannot establish a fact about the entire distribution. These edits never verify clinical outcome labels.

```
APPLY_REVIEWED_EVIDENCE = False
REVIEW_FILE = OUT / 'evidence_review_template.csv'
if APPLY_REVIEWED_EVIDENCE:
    REGISTRY = compat.apply_review_file(REGISTRY, REVIEW_FILE)
    print('Reviewed documentary facts applied. Rerun Sections 4–6 a
```

4. Run the metadata compatibility assessment

This section makes **55 pairwise comparisons** without loading any ECG data. The output distinguishes:

- **Known minimum effort:** Low, Moderate or High, based on explicit rules.
- **Evidence status:** unresolved facts prevent confirmation. A `?` on the matrix means unconfirmed, even when the known minimum is Low.
- **Clinical eligibility:** remains unverified for acute three-class supervision until appropriate clinical review.

The rules and missing fields are exported. High effort is not automatic exclusion, and Low is not permission to pool. A clinical reference standard, patient grouping, overlap review and later model validation are still required.

```
RESULTS = compat.run_study(REGISTRY, OUT, TARGET)
display(RESULTS['pairs'][['dataset_a', 'dataset_b', 'known_minimum_e
                          'evidence_status', 'evidence_coverage', 're
display(RESULTS['targets'])
```

```
Analyzing 11 documentary profiles; no ECG folders are accessed.
Writing comparison figures and clustering sensitivity analyses...
Complete: 55 pair assessments. Outputs: /content/ecg_metadata_compa
```

	dataset_a	dataset_b	known_minimum_effort	evidence_
0	ptb_xl	mimic_iv_ecg	Moderate	In
1	ptb_xl	ningbo	Low	In
2	ptb_xl	chapman_shaoxing	Low	In
3	ptb_xl	georgia	Moderate	In
				In

4	ptb_xl	cpsc_2018	Moderate	I
5	ptb_xl	cpsc_extra	Moderate	In
6	ptb_xl	sph	Moderate	In
7	ptb_xl	ptb_diagnostic	Moderate	In
8	ptb_xl	incart	High	In
9	ptb_xl	code15	Moderate	In
10	mimic_iv_ecg	ningbo	Moderate	In
11	mimic_iv_ecg	chapman_shaoxing	Moderate	In
12	mimic_iv_ecg	georgia	Moderate	In
13	mimic_iv_ecg	cpsc_2018	Moderate	In
14	mimic_iv_ecg	cpsc_extra	Moderate	In
15	mimic_iv_ecg	sph	Moderate	In
16	mimic_iv_ecg	ptb_diagnostic	Moderate	In
17	mimic_iv_ecg	incart	High	In
18	mimic_iv_ecg	code15	Moderate	In
19	ningbo	chapman_shaoxing	Low	In

	ningbo	chapman_shaoxing	Low	
20	ningbo	georgia	Moderate	In
21	ningbo	cpsc_2018	Moderate	In
22	ningbo	cpsc_extra	Moderate	In
23	ningbo	sph	Moderate	In
24	ningbo	ptb_diagnostic	Moderate	In
25	ningbo	incart	High	In
26	ningbo	code15	Moderate	In
27	chapman_shaoxing	georgia	Moderate	In
28	chapman_shaoxing	cpsc_2018	Moderate	In
29	chapman_shaoxing	cpsc_extra	Moderate	In
30	chapman_shaoxing	sph	Moderate	In
31	chapman_shaoxing	ptb_diagnostic	Moderate	In
32	chapman_shaoxing	incart	High	In
33	chapman_shaoxing	code15	Moderate	In
34	georgia	cpsc_2018	Moderate	In
35	georgia	cpsc_extra	Moderate	In
36	georgia	sph	Moderate	In

36	georgia	sph	Moderate	
37	georgia	ptb_diagnostic	Moderate	In
38	georgia	incart	High	In
39	georgia	code15	Moderate	In

5. Inspect structures and research figures

The structure guide shows documented layouts, not a scan of your folders. PNG and vector PDF versions of each figure are saved. The acquisition clusters are exploratory: unknown device/filter fields cannot be used to assert that clustered sources use the same equipment.

```
display(Markdown((OUT / 'DATASET_STRUCTURES.md').read_text()))
for name in ['dataset_attributes', 'evidence_completeness', 'pairwis
             'pairwise_evidence_coverage', 'harmonization_requiremen
    display(Image(filename=str(OUT / (name + '.png')), width=1050))
```

Selected dataset structures

These are documented example layouts, not an inventory of downloaded files.

PTB-XL — 1.0.3 / records500

```
ptb_xl/
  ptbxl_database.csv
  scp_statements.csv
  records500/00000/00001_hr.hea
  records500/00000/00001_hr.dat
  records100/00000/00001_lr.{hea,dat}
```

Use records500 for this profile. Do not count the 100 Hz copies as additional ECGs. Historical screenshot count 21,837 differs from release 1.0.3.

MIMIC-IV-ECG — 1.0

```
mimic_iv_ecg/
  record_list.csv
  machine_measurements.csv
  machine_measurements_data_dictionary.csv
  waveform_note_links.csv
```

```
  files/pXXXX/pSUBJECT/sSTUDY/STUDY.{hea,dat}
```

bandwidth and filtering fields can describe within-source heterogeneity. No device model is inferred from cart_id.

Ningbo — Challenge 2021 / 1.0.3

```
ningbo/
  [optional subfolders]/
    RECORD_ID.hea
    RECORD_ID.mat  # val: leads x samples
```

Counts refer to public challenge subsets. Device information remains unknown; GE MUSE is not proof of an acquisition model.

Chapman-Shaoxing — Challenge 2021 / 1.0.3

```
chapman_shaoxing/
  [optional subfolders]/
    RECORD_ID.hea
    RECORD_ID.mat  # val: leads x samples
```

Counts refer to public challenge subsets. Device information remains unknown; GE MUSE is not proof of an acquisition model.

Georgia 12-lead — Challenge 2021 / 1.0.3

```
georgia/
  [optional subfolders]/
    RECORD_ID.hea
    RECORD_ID.mat  # val: leads x samples
```

Collection year must not be inferred from the challenge name. CPSC duration range is documented for the combined source.

CPSC 2018 — Challenge 2021 / 1.0.3

```
cpsc_2018/
  [optional subfolders]/
    RECORD_ID.hea
    RECORD_ID.mat  # val: leads x samples
```

Collection year must not be inferred from the challenge name. CPSC duration range is documented for the combined source.

CPSC-Extra — Challenge 2021 / 1.0.3

```
cpsc_extra/
  [optional subfolders]/
    RECORD_ID.hea
    RECORD_ID.mat  # val: leads x samples
```

Collection year must not be inferred from the challenge name. CPSC duration range is documented for the combined source.

Shandong Provincial Hospital — Figshare collection 5779802 v1

```
sph/
  metadata.csv
  code.csv
  [waveform folder]/RECORD_ID.h5  # 12 x variable samples
```

PTB Diagnostic — 1.0.0 native

```
ptb_diagnostic/
  patient001/
    s0010_re.hea
    s0010_re.dat  # standard twelve + Frank X/Y/Z
```

PTB page reports inconsistent subject counts in its announcement and detailed description; no patient total is asserted here. Recorder bandwidth is not entered as a digital low-pass cutoff.

INCART — 1.0.0 native

```
incart/
  I01.hea
  I01.dat
  I01.atr  # beat annotations
```

CODE-15% — Zenodo 4916206

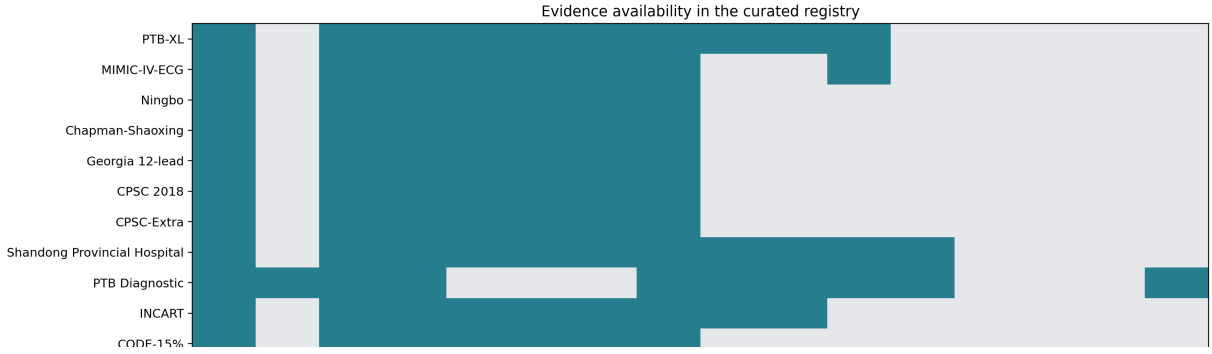
```
code15/
  exams.csv
  exams_part0.hdf5
    exam_id: (N,)
    tracings: (N, 4096, 12)
  exams_part1.hdf5 ...
```

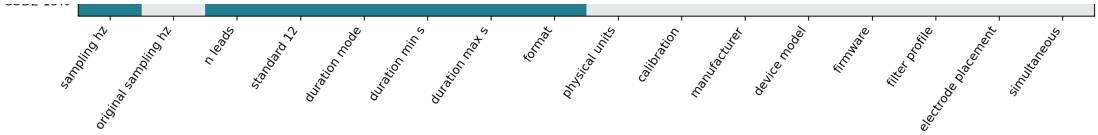
4096/400 is stored duration including padding, not measured ECG duration. Amplitude calibration is deliberately unresolved.

Selected-distribution attributes (publisher documentation)

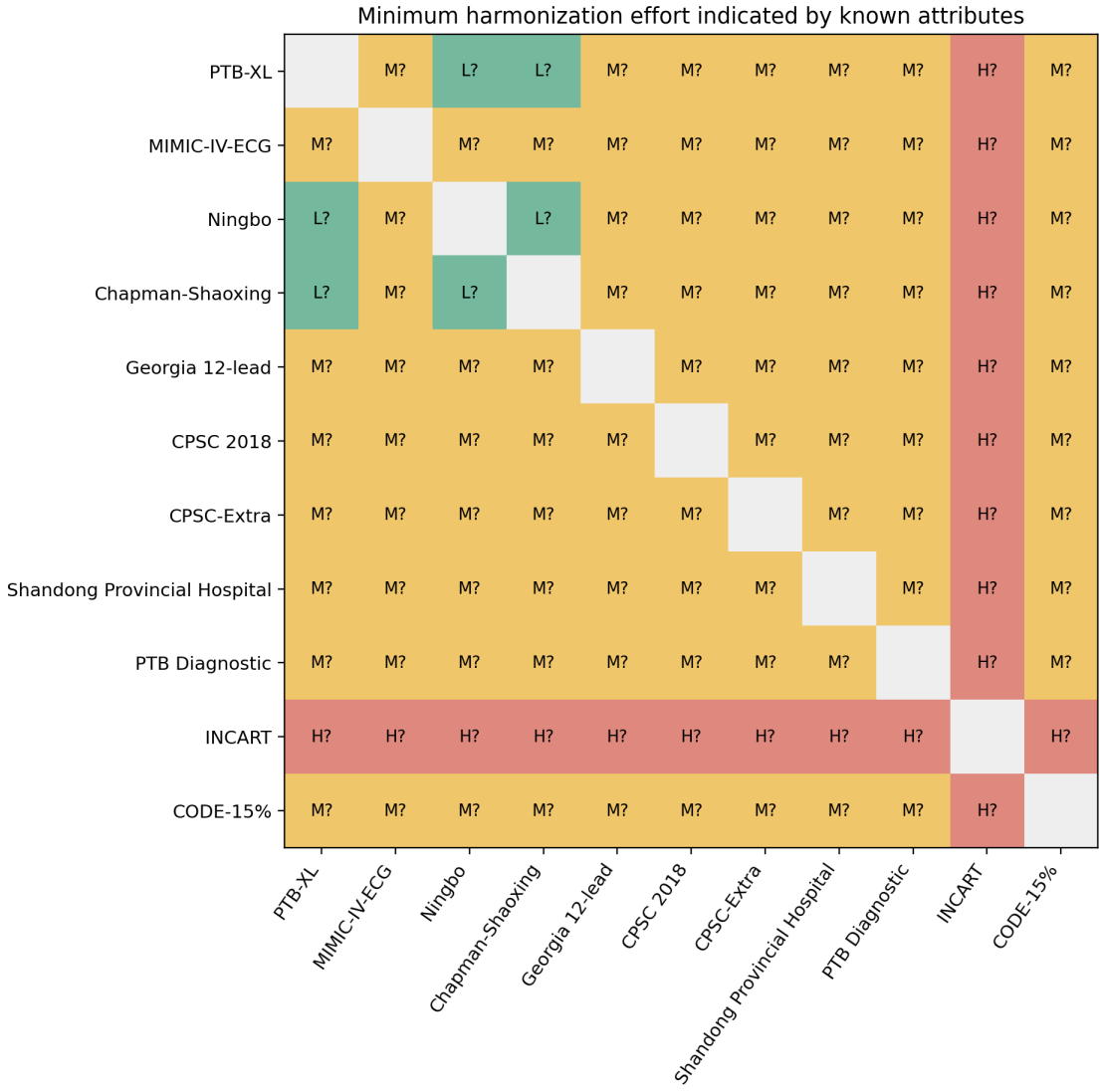
	Rate (Hz)	Channels	Min duration (s)	Max duration (s)	Container	Manufacturer
PTB-XL	500	12	10	10	WFDB-DAT	Schiller AG
MIMIC-IV-ECG	500	12	10	10	WFDB-DAT	Burdick/Spacelabs; Philips; GE
Ningbo	500	12	10	10	WFDB-MAT	Unknown
Chapman-Shaoxing	500	12	10	10	WFDB-MAT	Unknown
Georgia 12-lead	500	12	5	10	WFDB-MAT	Unknown
CPSC 2018	500	12	6	144	WFDB-MAT	Unknown
CPSC-Extra	500	12	6	144	WFDB-MAT	Unknown
Shandong Provincial Hospital	500	12	10	60	HDF5-record	MedEx
PTB Diagnostic	1000	15	Unknown	Unknown	WFDB-DAT	PTB
INCART	257	12	1800	1800	WFDB-DAT	Unknown
CODE-15%	400	12	7	10	HDF5-shard	Unknown

Rates describe distributed files. Duration excludes padding where documented. Unknown values are not imputed.

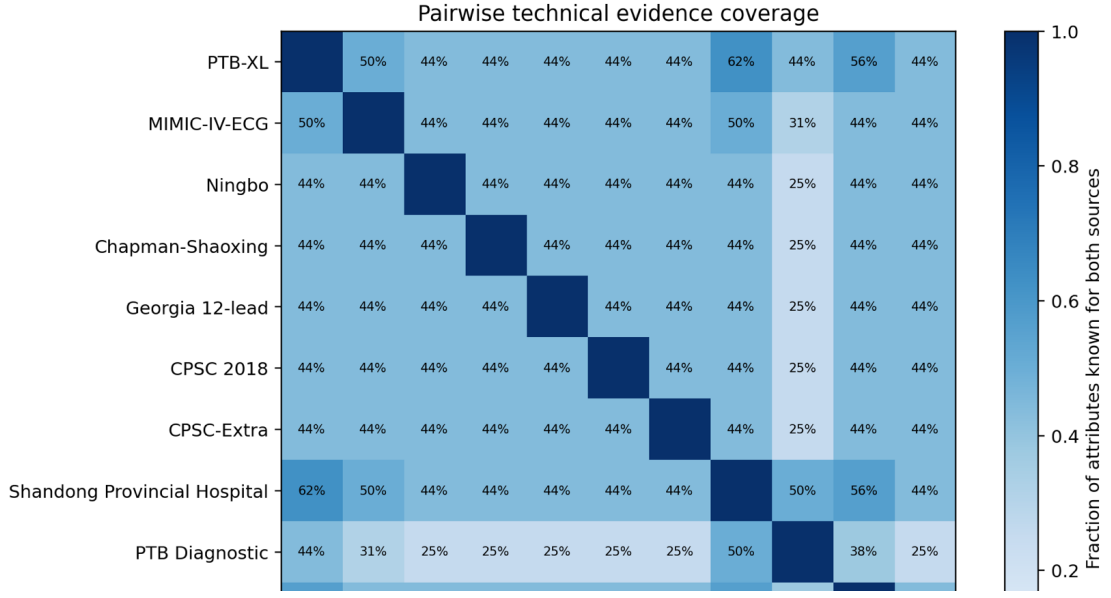


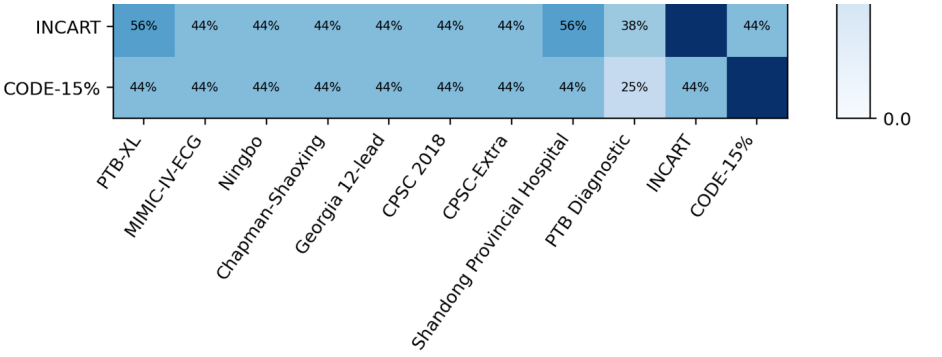


Teal = documented value; gray = unknown in this evidence set. Availability does not certify uniformity or accuracy.

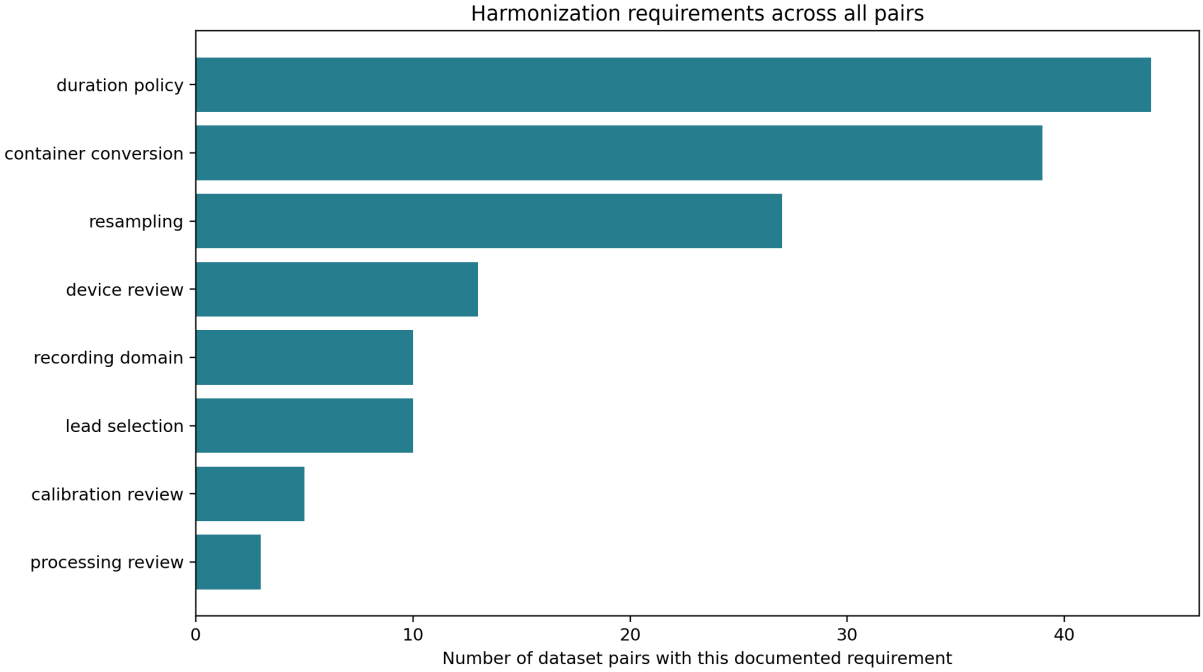


L = Low, M = Moderate, H = High. ? = insufficient evidence to confirm the tier. These are not pooling approvals.

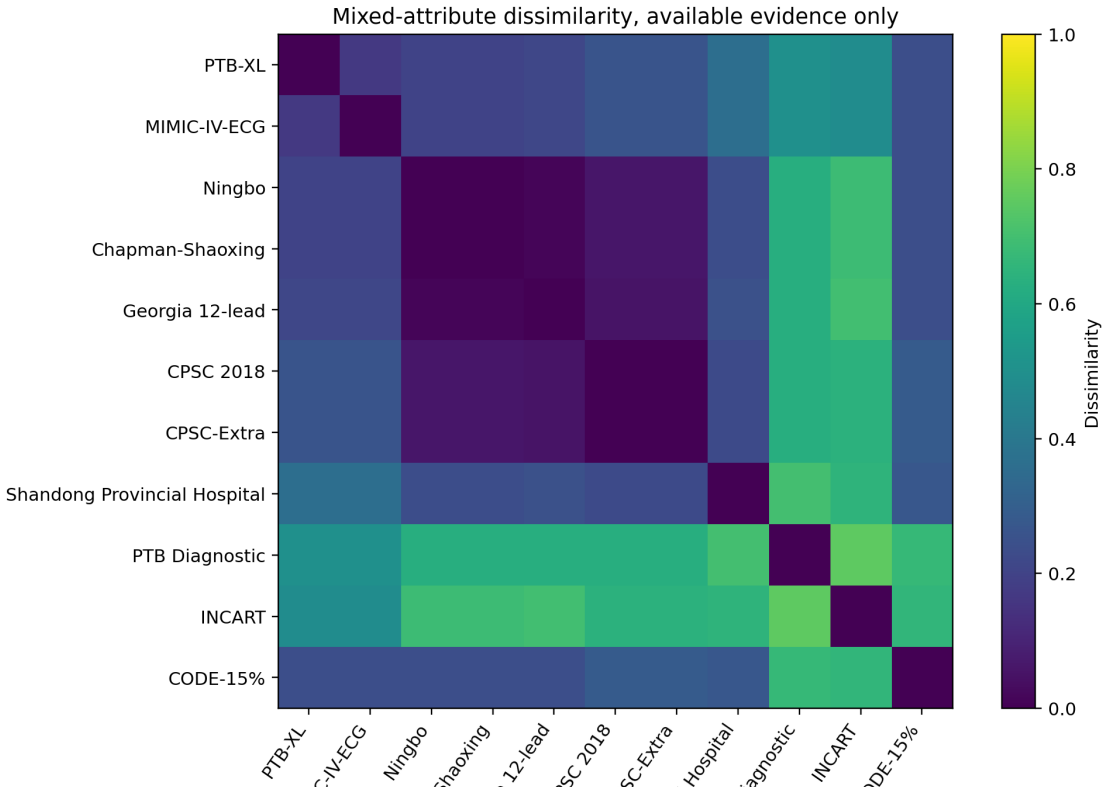


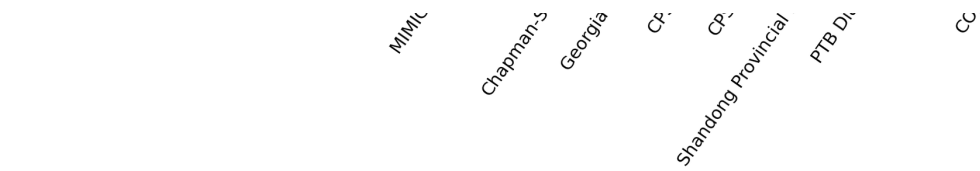


Diagonal is not a self-validation result. Coverage is not a probability of compatibility.



Multiple requirements can apply to the same pair; unknown requirements are not counted as absent.





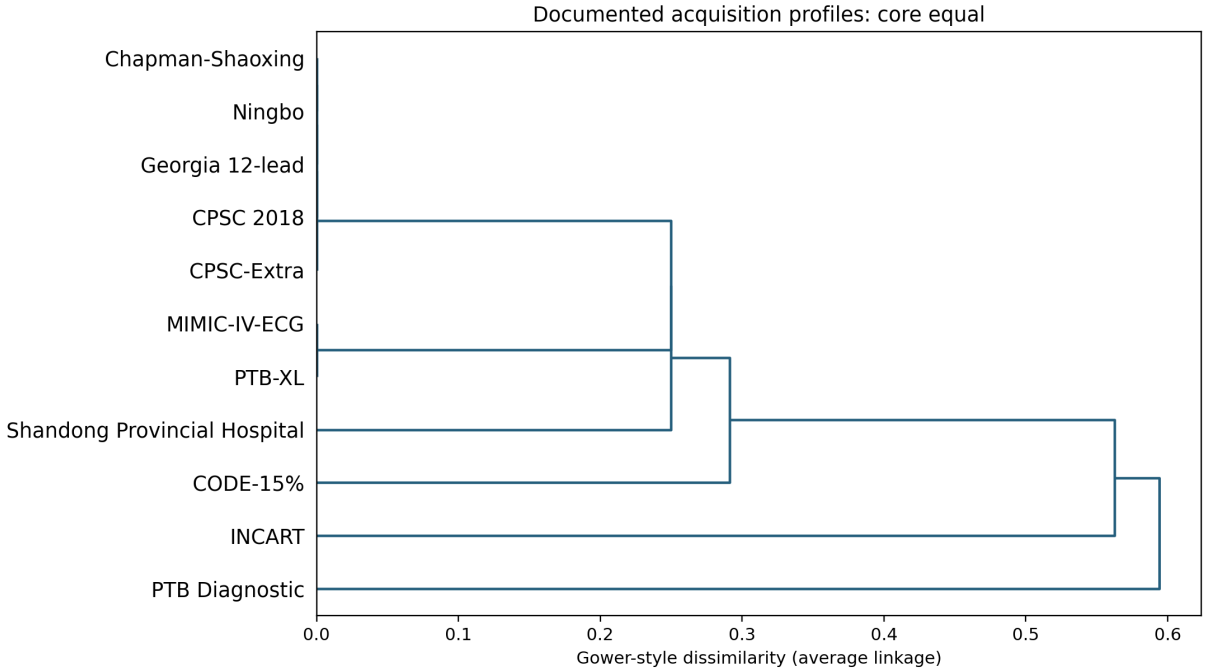
Gray = insufficient comparable weight. Interpret together with mixed_distance_coverage.csv; no missing-value imputation.

Clustering and sensitivity analysis

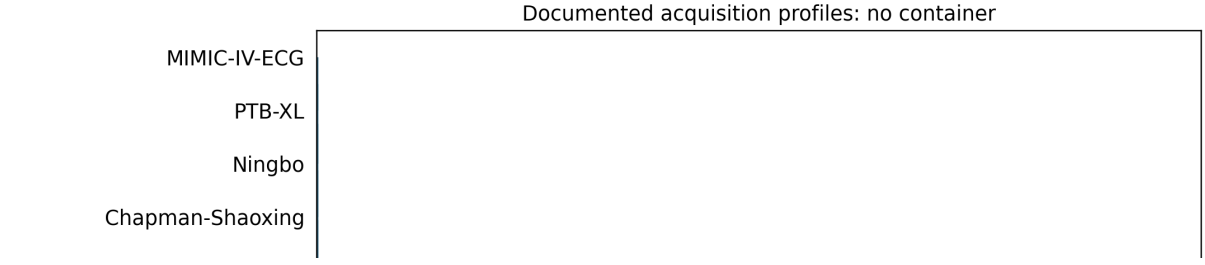
Primary clustering uses the fully observed subset of four prespecified attributes: distributed sampling rate, channel count, container and diagnostic/Holter context. It uses a transparent Gower-style dissimilarity and average linkage. It does not force three clusters or use patient-level ECG features.

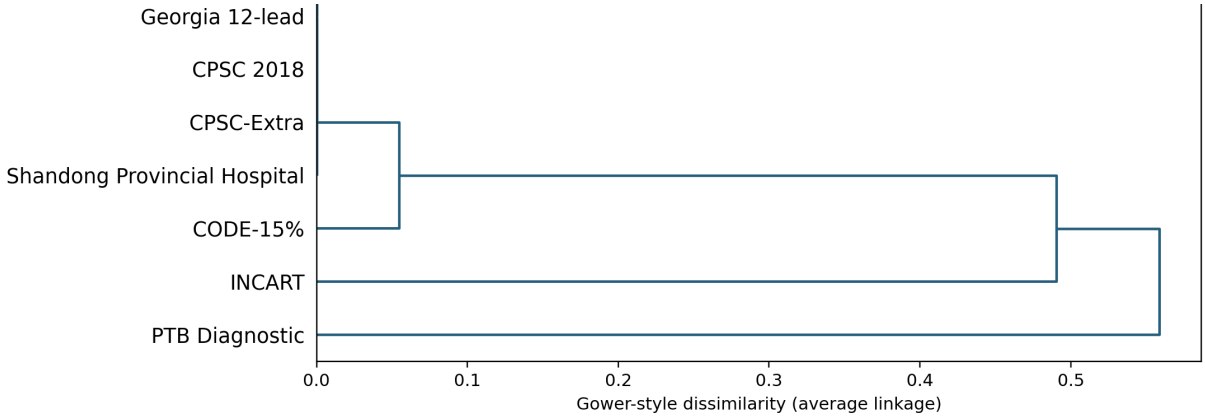
Two additional trees remove container and reduce its weight. Similarity caused by a common file format is not acquisition equivalence. Exact feature sets, normalizations, distances and linkage matrices are exported. The broader pairwise mixed-attribute matrix omits missing fields, reports coverage, and is not used to construct the trees.

```
for name in ['core_equal', 'no_container', 'acquisition_weighted']:
    display(Image(filename=str(OUT / ('clustering_' + name + '.png')
display(RESULTS['sensitivity'])
```

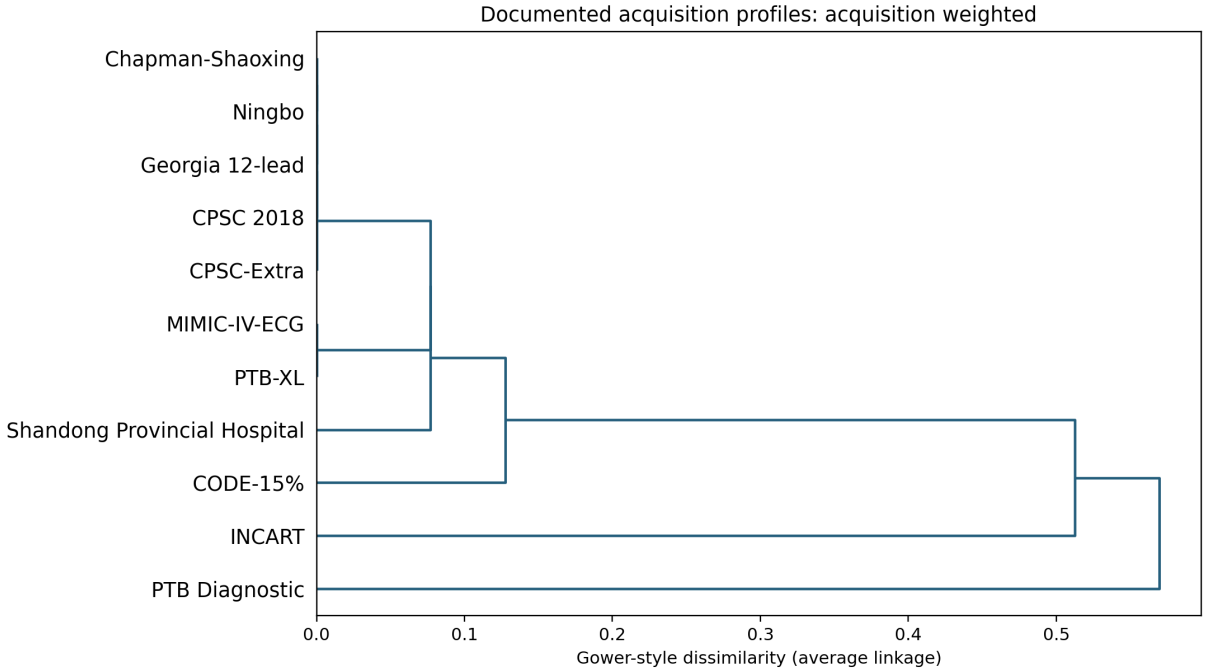


Dataset-level descriptors only. Identical profiles can form zero-distance ties; missing device/filter fields are excluded.





Dataset-level descriptors only. Identical profiles can form zero-distance ties; missing device/filter fields are excluded.



Dataset-level descriptors only. Identical profiles can form zero-distance ties; missing device/filter fields are excluded.

	scenario	distance_spearman	cophenetic_spearman	mean_abs
0	core_equal	1.000000	1.000000	
1	no_container	0.937122	0.948954	
2	acquisition_weighted	1.000000	0.988883	

6. Read the methods, limitations and handoff tables

This is the end of the required study. The next sections are optional. Do not report this as waveform-quality testing or as evidence that a model generalizes across sources. Do not convert absent MI labels to non-MI. PTB-XL MI statements, machine interpretation text, AHA/SNOMED codes and CODE rhythm labels require separate clinical review.

```
display(Markdown((OUT / 'METHODS_AND_RESULTS.md').read_text()))
PAIR_FILTER = None  # Example: 'ptb_xl'. Leave None to inspect all
pairs = RESULTS['pairs']
if PAIR_FILTER:
    pairs = pairs[(pairs.dataset_a == PAIR_FILTER) | (pairs.dataset
display(pairs[['dataset_a', 'dataset_b', 'known_minimum_effort', 'u
```

Metadata-based ECG dataset compatibility assessment

Scope and design

This run compared 11 version-specific documentary dataset profiles and 55 unordered pairs. It assessed acquisition/representation compatibility for ECG-only acute MI research. No ECG sample arrays were read, no signal quality or morphology was measured, and no STEMI/NSTEMI/non-MI labels were generated. The registry curation date is 2026-10-10. Publisher counts are not observed local sample sizes. The complete run specification, engine version, packages and registry digest are in run_provenance.json.

Evidence and decisions

Each field has a value, source pointer, locator, scope and evidence status in attribute_evidence.csv. Unknown means not established in this evidence set. It does not mean a property is absent or identical. The highest applicable rule determines the minimum effort indicated by known attributes. This is a study-defined operational rule, not a validated clinical scale. Critical and additional unresolved fields prevent confirmation of an effort tier. Device equality is not established from brand or cart identifier alone. Numeric evidence coverage is a completeness measure, not calibrated confidence. Technical readiness and clinical eligibility are separate. No source is approved for supervised acute three-class training by this assessment.

Descriptive results from documentation

Known minimum effort counts: {"Moderate": 42, "High": 10, "Low": 3}. Unconfirmed pairs: 55. These counts describe the rules applied to the registry; they are not experimental pooling results. Use pairwise_assessment.csv and pairwise_actions.csv to inspect every decision. Dataset-level target preparation is reported separately in target_readiness.csv. Figure captions explain the question-mark notation.

Dissimilarity and clustering

Available-field Gower-style dissimilarities combine normalized numeric differences, categorical mismatch and Jaccard distance for manufacturer sets. Sampling frequency and duration are log2-transformed. Numeric ranges are defined over this registry and nonvarying features are excluded. Missing fields are omitted from each pair's weighted denominator. Pairs below 60% comparable eligible weight are masked. These available-field distances are NOT fed to hierarchical clustering because pairs can be compared on different attributes. Hierarchical clustering instead uses the fully observed subset of the prespecified core features: distributed sampling frequency, channel count, container and recording context, with average linkage. The exact retained features are saved in clustering_specification.json. Device/filter similarity cannot be concluded from these clusters because those fields lack adequate evidence. Equal-profile ties are possible; no three-cluster result is forced. Two sensitivity analyses remove container and downweight container relative to acquisition features. Distance and cophenetic Spearman correlations quantify agreement with the baseline tree descriptively. No p-values or patient-level inference are claimed for eleven documentary dataset profiles.

Optional local observations

CSV summaries, bounded header checks and HDF5 schema inspections are separate artifacts. They never silently replace documentary facts or certify a full download. Preview rows are a convenience subset, not a random population sample. Header selection is explicitly bounded and may miss configurations, especially in a partial download. Only a complete metadata-table pass may describe frequencies for that table; it still does not prove waveform completeness. No waveform hashes or near-duplicate signal detection are performed. Population shift, label validity, signal quality and cross-dataset generalization require subsequent studies.

Use in a manuscript

Describe this as a documentary and optional metadata feasibility assessment. Report the edition, input target, evidence coverage, rules, optional sample scope, and clustering sensitivity. Review missing device/filter facts and clinical labels with domain experts. Keep patients and overlapping exports separated across train/test splits. Do not state that this notebook validates acute MI labels or demonstrates model performance.

	dataset_a	dataset_b	known_minimum_effort	unresolv
0	ptb_xl	mimic_iv_ecg	Moderate	physical_ d
1	ptb_xl	ningbo	Low	physical_ ma
2	ptb_xl	chapman_shaoxing	Low	physical_ ma
3	ptb_xl	georgia	Moderate	physical_ ma
4	ptb_xl	cpsc_2018	Moderate	physical_ ma
5	ptb_xl	cpsc_extra	Moderate	physical_ ma
6	ptb_xl	sph	Moderate	device_mo ele

7	ptb_xl	ptb_diagnostic	Moderate	device_mo ele
8	ptb_xl	incart	High	device_mo
9	ptb_xl	code15	Moderate	physical_ ma
10	mimic_iv_ecg	ningbo	Moderate	physical_ ma
11	mimic_iv_ecg	chapman_shaoxing	Moderate	physical_ ma
12	mimic_iv_ecg	georgia	Moderate	physical_ ma
13	mimic_iv_ecg	cpsc_2018	Moderate	physical_ ma
14	mimic_iv_ecg	cpsc_extra	Moderate	physical_ ma
15	mimic_iv_ecg	sph	Moderate	physical_ d

7. Optional: inspect existing metadata tables only

Disabled by default. Set `RUN_LOCAL_METADATA=True` to mount Drive and read explicit metadata files. No waveform folder is traversed. Missing tables are reported and skipped; there is no download.

`MAX_METADATA_ROWS=25000` is a convenience prefix preview, not a representative sample. Set it to `None` only when you need complete-table frequencies. A full CSV pass reads selected columns in chunks and saves a cache keyed by path, file size, modification time and settings. It may still take time on Drive. The cache does not constitute checksum validation.

MIMIC `cart_id`, `bandwidth` and `filtering` are summarized as released categories. A cart ID is not a manufacturer/model. Raw filter strings are not silently interpreted as numeric cutoff frequencies. No machine-derived ECG measurements are selected by default.

```
RUN_LOCAL_METADATA = False
MAX_METADATA_ROWS = 25000  # None = entire metadata table; no wavef
DATA_ROOT = Path('/content/drive/MyDrive/MI_research/data')
LOCAL_OUT = OUT / 'local_metadata'
LOCAL_SUMMARIES = {}
if RUN_LOCAL_METADATA:
    from google.colab import drive
    drive.mount('/content/drive')
    files_to_summarize = {
        'ptb_xl': DATA_ROOT / 'ptb_xl' / 'ptbxl_database.csv',
        'mimic_iv_ecg': DATA_ROOT / 'mimic_iv_ecg' / 'machine_measu
        'sph': DATA_ROOT / 'sph' / 'metadata.csv',
        'code15': DATA_ROOT / 'code15' / 'exams.csv',
    }
    for dataset, path in files_to_summarize.items():
        summary = compat.csv_summary(path, LOCAL_OUT, max_rows=MAX_
        LOCAL_SUMMARIES[dataset] = summary
        print(dataset, summary['status'], summary.get('scope', ''),
        print('Columns:', summary.get('schema', []))
        for image_path in compat.plot_csv_summary(summary, LOCAL_OU
            display(Image(filename=image_path, width=900))
else:
    print('Local metadata inspection skipped. Documentary results a
```

```
Local metadata inspection skipped. Documentary results are complete.
```

8. Optional: validate a bounded set of local headers

Disabled by default. This reads only selected `.hea` files and never opens waveform samples. The default selects up to 50 candidates per source from the first 10,000 metadata rows with a fixed seed. Missing local headers are recorded; the code does not search for replacements or scan folders. In a partial MIMIC download, many candidates may be missing.

For known files, use `EXPLICIT_HEADERS` with relative `.hea` paths. For a reviewed stratified candidate table, call `select_header_paths(..., stratum_column='your_column')`; the table must contain both paths and the stratum. This is coverage-oriented selection, not a full-release prevalence estimate. Sampling logs, observed gain/units/lead order and disagreements with documented frequency/channel/sample counts are exported separately. They do not silently change the registry.

```
RUN_HEADER_CHECK = False
HEADERS_PER_DATASET = 50
EXPLICIT_HEADERS = {
    # 'ptb_xl': ['records500/00000/00001_hr.hea'],
    # 'mimic_iv_ecg': ['files/pXXXX/pSUBJECT/sSTUDY/STUDY.hea'],
}
HEADER_OUT = OUT / 'header_checks'
if RUN_HEADER_CHECK:
    from google.colab import drive
    drive.mount('/content/drive')
    if importlib.util.find_spec('wfdb') is None:
        subprocess.check_call([sys.executable, '-m', 'pip', 'instal
    candidates = {'ptb_xl': ('ptbxl_database.csv', 'filename_hr'),
                  'mimic_iv_ecg': ('record_list.csv', 'path')}
    by_id = {d['id']: d for d in REGISTRY['datasets']}
    for dataset in sorted(set(candidates) | set(EXPLICIT_HEADERS)):
        root = DATA_ROOT / dataset
        if dataset in EXPLICIT_HEADERS:
            selected = EXPLICIT_HEADERS[dataset]
            selection = {'scope': 'User-specified convenience sampl
        else:
            filename, column = candidates[dataset]
            if not (root / filename).is_file():
                print(dataset, 'metadata table missing; skipped')
                continue
            selected, selection = compat.select_header_paths(root /
        expected = {f: compat.val(by_id[dataset], f) for f in ['sam
        observed = compat.inspect_headers(root, selected, HEADER_OU
                                          expected=expected, limit=
        display(observed)
        compat.plot_header_observations(observed, HEADER_OUT, datas
else:
    print('Header checks skipped; no local ECG files accessed.')
```

```
Header checks skipped; no local ECG files accessed.
```

9. Optional: inspect HDF5 structure without reading samples

For SPH or CODE, supply one explicit file and the documented dataset keys. The function inspects shape, dtype, chunking and compression only; it does not slice arrays or load `exam_id` values. It does not follow external HDF5 links. Schema observations are not an amplitude-calibration test.

```
RUN_HDF5_SCHEMA = False
HDF5_PATH = DATA_ROOT / 'code15' / 'exams_part0.hdf5'
HDF5_KEYS = ['tracings', 'exam_id']  # For SPH, supply the actual d
if RUN_HDF5_SCHEMA:
    from google.colab import drive
    drive.mount('/content/drive')
    if importlib.util.find_spec('h5py') is None:
        subprocess.check_call([sys.executable, '-m', 'pip', 'instal
    display(compat.inspect_hdf5(HDF5_PATH, OUT / 'hdf5_schema.csv',
else:
    print('HDF5 inspection skipped.')
```

```
HDF5 inspection skipped.
```

10. Export the research package

Creates one ZIP on the Colab runtime containing the tables, figures, methods, source register, exact engine and registry used. Set `SAVE_ZIP_TO_DRIVE=True` to preserve it across runtime deletion, or `DOWNLOAD_ZIP=True` to download it to your computer. Saving results does not require copying any ECG files. Optional header outputs contain record paths; review them before public sharing.

```
import shutil
SAVE_ZIP_TO_DRIVE = False
DOWNLOAD_ZIP = False
for filename in ['ecg_metadata_compatibility.py', 'dataset_registry
    shutil.copy2(ENGINE_DIR / filename, OUT / filename)
archive = Path(shutil.make_archive(str(WORK / RUN_NAME), 'zip', OUT
print('Results ZIP:', archive)
if SAVE_ZIP_TO_DRIVE:
    from google.colab import drive
    drive.mount('/content/drive')
    destination = Path('/content/drive/MyDrive/MI_research/metadata
    destination.mkdir(parents=True, exist_ok=True)
    shutil.copy2(archive, destination / archive.name)
    print('Saved to Drive:', destination / archive.name)
if DOWNLOAD_ZIP:
    from google.colab import files
    files.download(str(archive))
```

```
Results ZIP: /content/ecg_metadata_compatibility/metadata_compatibil
```

11. Reconnect and interpret correctly

If only the browser disconnected: reconnect and check whether the runtime and variables remain. Do not launch a second copy of a running cell.

If Colab created a new runtime: rerun Sections 1–6. The documentary assessment is small and requires no dataset inventory or repeated downloads. If you saved the results ZIP in Section 10, the earlier results remain on Drive. Optional metadata caches are inside that ZIP; restore them to the same output folder to reuse summaries when the input file size/mtime and settings still match. You may instead leave Sections 7–9 disabled.

When handing off to the AI team: provide `target_readiness.csv`, `pairwise_assessment.csv`, `pairwise_actions.csv`, `attribute_evidence.csv`, `METHODS_AND_RESULTS.md`, and the figure PDFs. Explain that technical similarity, clinical label validity and model generalization are separate questions. A source with uncertain labels may be a pretraining candidate after review, rather than an automatically valid supervised acute-MI cohort.

Recommended wording: “We assessed documented acquisition and representation characteristics using prespecified harmonization rules. We retained unknown attributes, quantified evidence coverage and performed exploratory dataset-level clustering with sensitivity analyses. This assessment did not evaluate waveform quality or diagnostic performance.”